

EDC BLOCK-003 Derivation v13

Weak-Field 5D→4D Matching:
The Normalization Extractor

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Abstract

Status: Program note; not closure.

This note establishes the functional form by which the 4D Newton constant G_N emerges from a 5D gravitational theory with a compact or warped extra dimension. We derive the *normalization extractor*: an explicit integral $\mathcal{I}$ such that $M_{\text{Pl}}^2 = M_5^3 \times \mathcal{I}$, from which $G_N = 1/(8\pi M_{\text{Pl}}^2)$. The derivation is standard (Randall–Sundrum, Kaluza–Klein), imported here to identify precisely where EDC must supply physical input.

One-line outcome:

BRIDGE SLOT FOUND: G_N reduces to $M_5^3 \mathcal{I}$; EDC must compute $\mathcal{I}$ or provide one calibration scale.

1 Introduction

The BLOCK-003 program seeks to derive the 4D Newton constant G_N from EDC’s 5D brane-world structure. Previous notes (v1–v12) established:

- The standard result $G_N = \kappa_5^2/(6\pi L)$ for flat compactification [I]
- The constant C in $\kappa_5^2 = C \cdot \sigma^{-3/4}$ remains unfixed
- Part I of EDC imports 4D gravity as [I]/[P], providing no 5D derivation

This note takes a different approach: we derive the *general structure* of the 5D→4D reduction in the weak-field (Newtonian) limit, making explicit the normalization integral that converts 5D to 4D gravitational coupling. This identifies the precise “bridge slot” that EDC must fill.

2 Setup: 5D Action and Linearization

2.1 The 5D Action

We consider the 5D Einstein–Hilbert action with Gibbons–Hawking–York boundary term and brane-localized stress-energy [I]:

$$S = \frac{1}{2\kappa_5^2} \int d^5x \sqrt{-g^{(5)}} R^{(5)} + \frac{1}{\kappa_5^2} \int_{\partial\mathcal{M}} d^4x \sqrt{-h} K - \int_{\text{brane}} d^4x \sqrt{-\gamma} \sigma \quad (1)$$

where:

- $\kappa_5^2 = 1/M_5^3$ is the 5D gravitational coupling [I]
- K is the extrinsic curvature of the boundary [M]
- σ is the brane tension [P]

2.2 Linearization Ansatz

For weak-field analysis, we write the 5D metric as [M]:

$$ds^2 = e^{2A(\xi)} (\eta_{\mu\nu} + h_{\mu\nu}(x, \xi)) dx^\mu dx^\nu + d\xi^2 \quad (2)$$

where $A(\xi)$ is the warp factor (with $A = 0$ for flat compactification) and $h_{\mu\nu}$ is a small perturbation sourced by a 4D point mass M on the brane.

3 Non-Compact Extra Dimension: The $1/r^2$ Failure

Why non-compact gives $1/r^2$: In a non-compact 5D spacetime without warping, the gravitational potential of a point mass propagates in all five dimensions [D]. The Green's function for the 5D Laplacian scales as [M]:

$$G^{(5)}(\vec{r}, \xi) \sim \frac{1}{(r^2 + \xi^2)^{3/2}} \quad (3)$$

Evaluating at the brane location $\xi = 0$, or integrating over unbounded ξ , yields $h_{00} \sim 1/r^2$ rather than $1/r$ [D].

This violates the observed $1/r$ Newtonian potential. The non-compact case is therefore **excluded by observation** [BL].

Recovery requires either:

1. Compact extra dimension (finite ξ range), or
2. Warped geometry that localizes gravity to the brane

4 Compact/Warped Recovery of $1/r$

4.1 Kaluza–Klein Decomposition

For a compact or warped extra dimension, we decompose the 4D graviton perturbation in modes [I]:

$$h_{\mu\nu}(x, \xi) = \sum_n h_{\mu\nu}^{(n)}(x) \psi_n(\xi) \quad (4)$$

where $\psi_n(\xi)$ satisfies the eigenvalue equation [D]:

$$-\frac{1}{w(\xi)} \frac{d}{d\xi} \left(w(\xi) \frac{d\psi_n}{d\xi} \right) = m_n^2 \psi_n \quad (5)$$

with weight function $w(\xi) = e^{4A(\xi)}$ and appropriate boundary conditions.

4.2 The Zero-Mode

The zero-mode ($n = 0$, $m_0 = 0$) is the 4D massless graviton. Its profile $\psi_0(\xi)$ must be **normalizable** with respect to the weighted inner product [D]:

$$\boxed{\int d\xi w(\xi) |\psi_0(\xi)|^2 = 1} \quad (6)$$

Key insight: When a normalizable zero-mode exists, the 4D effective theory contains a massless spin-2 field (the graviton) that mediates $1/r$ gravity [D]. The massive KK modes ($n \geq 1$) contribute only Yukawa-suppressed corrections $\sim e^{-m_n r}/r$ [D].

4.3 Conditions for Normalizability

1. **Compact flat:** $A(\xi) = 0$, $\xi \in [0, L]$. Then $w = 1$ and $\psi_0 = \text{const}$. Normalizability requires finite L [D].
2. **Warped (RS-type):** $A(\xi) = -k|\xi|$ with $\xi \in [-L, L]$ or orbifold $S^1/\mathbb{Z}_2$. The weight $w = e^{-4k|\xi|}$ localizes the zero-mode near $\xi = 0$. Normalizability holds even for $L \rightarrow \infty$ if the warp factor decays sufficiently [I].

5 The Normalization Extractor

This is the core result: the explicit relationship between 5D and 4D gravitational couplings.

5.1 Derivation

Substituting the KK decomposition (4) into the 5D action (1) and integrating over ξ , the 4D effective action for the zero-mode becomes [D]:

$$S_{\text{eff}}^{(4)} = \frac{1}{2\kappa_5^2} \int d^4x \sqrt{-g^{(4)}} R^{(4)} \underbrace{\int d\xi w(\xi) |\psi_0(\xi)|^2}_{=\mathcal{I}/M_5^{-3}} \quad (7)$$

Comparing with the standard 4D Einstein–Hilbert action [M]:

$$S^{(4)} = \frac{M_{\text{Pl}}^2}{2} \int d^4x \sqrt{-g^{(4)}} R^{(4)} \quad (8)$$

we obtain the **normalization extractor**:

$$\boxed{M_{\text{Pl}}^2 = M_5^3 \times \mathcal{I} \quad \text{where} \quad \mathcal{I} \equiv \int d\xi e^{4A(\xi)} |\psi_0(\xi)|^2} \quad (9)$$

The 4D Newton constant follows from [M]:

$$G_N = \frac{1}{8\pi M_{\text{Pl}}^2} = \frac{1}{8\pi M_5^3 \mathcal{I}} \quad (10)$$

5.2 Explicit Forms

Flat compactification ($A = 0$, $\xi \in [0, L]$): With $\psi_0 = 1/\sqrt{L}$ for normalization [D]:

$$\mathcal{I}_{\text{flat}} = \int_0^L d\xi \cdot 1 \cdot \frac{1}{L} = 1 \quad \Rightarrow \quad M_{\text{Pl}}^2 = M_5^3 L \quad (11)$$

Hence $G_N = 1/(8\pi M_5^3 L)$, which matches v1 result with $\kappa_5^2 = M_5^{-3}$ [D].

RS-type warping ($A = -k|\xi|$, **orbifold**): With the zero-mode profile $\psi_0(\xi) \propto e^{2k|\xi|}$ (localized near UV brane) and proper normalization [I]:

$$\mathcal{I}_{\text{RS}} = \int_{-L}^L d\xi e^{-4k|\xi|} \cdot e^{4k|\xi|} = 2L \quad \Rightarrow \quad M_{\text{Pl}}^2 = 2M_5^3 L \quad (12)$$

(For the infinite RS-II limit with one brane, $\mathcal{I}_{\text{RS-II}} = 1/k$.)

5.3 The Bridge Slot

Equation (9) makes explicit what is needed to compute G_N :

Bridge Slot: EDC must provide one of the following:

1. The warp factor $A(\xi)$ and integration domain (compactification scale L)
2. The 5D Planck mass M_5 from first principles
3. One measured scale (e.g., G_N^{obs}) to calibrate the integral $\mathcal{I}$

Without at least one of these, G_N cannot be predicted.

6 Schematic Figure

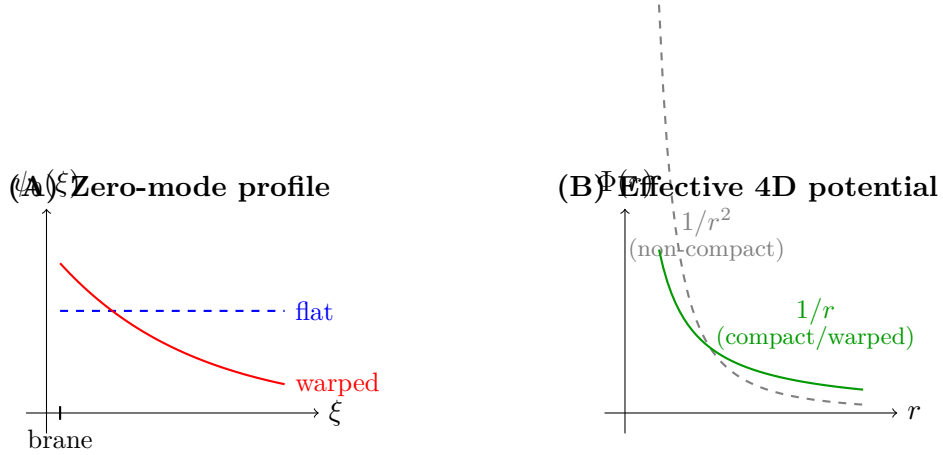


Figure 1: **(A)** Zero-mode profile $\psi_0(\xi)$: constant for flat compactification, localized near brane for warped geometry. **(B)** Effective 4D gravitational potential: non-compact gives $1/r^2$ (excluded by observation), compact/warped recovers $1/r$ (Newtonian). *Schematic only, not to scale.*

7 Epistemic Classification Table

Legend: [M] Mathematical identity; [D] Derived within this note; [Dc] Derived conditional on stated assumption; [P] Postulate; [I] Imported from literature; [BL] Baseline (observational input); [Cal] Calibrated to observation.

8 Conclusion

8.1 What v13 Closes

1. **Functional form:** The relationship $G_N = 1/(8\pi M_5^3 \mathcal{I})$ is now explicit.
2. **Normalization slot:** The integral $\mathcal{I} = \int d\xi e^{4A} |\psi_0|^2$ is identified as the key quantity linking 5D and 4D.
3. **Exclusion:** Non-compact extra dimensions are excluded by observation ($1/r^2$ vs $1/r$).
4. **Recovery mechanism:** Compact or warped geometries restore 4D gravity via a normalizable zero-mode.

Table 1: Epistemic ledger for derivation v13

Element	Tag	Source/Status
5D EH action structure	[I]	Imported from GR/RS literature
GHY boundary term	[I]	York (1972), Gibbons–Hawking (1977)
Brane tension term	[P]	Standard RS ansatz
Linearization ansatz	[M]	Mathematical expansion
5D Green’s function $\sim 1/r^2$	[D]	Derived from 5D Laplacian
Non-compact $\Rightarrow$ excluded	[BL]	Observation: $\Phi \propto 1/r$
KK decomposition	[I]	Kaluza (1921), Klein (1926)
Eigenvalue equation for ψ_n	[D]	Derived from variation
Normalization condition	[M]	Definition of unit norm
Zero-mode $\Rightarrow 1/r$	[D]	Derived from 4D effective theory
$M_{\text{Pl}}^2 = M_5^3 \mathcal{I}$	[D]	Derived (dimensional reduction)
$G_N = 1/(8\pi M_{\text{Pl}}^2)$	[M]	Definition
Flat case: $\mathcal{I} = L$	[D]	Derived (explicit integration)
RS case: $\mathcal{I} = 2L$ or $1/k$	[I]	RS model (1999)
Warp factor $A(\xi)$	[P/I]	Must be specified by EDC
M_5 value	[P/Cal]	Postulate or calibration

8.2 What Remains Open

For EDC to close BLOCK-003, it must provide:

Gap	Required Input	Status
Warp factor	$A(\xi)$ from EDC dynamics	OPEN
Compactification scale	L or equivalent	OPEN
5D Planck mass	M_5 from first principles	OPEN
One calibration	G_N^{obs} or ℓ_P^{obs}	Available [BL]

8.3 One-Line Outcome

BRIDGE SLOT FOUND: G_N reduces to $M_5^3 \mathcal{I}$;
EDC must compute $\mathcal{I}$ or provide one calibration scale.

References

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